

ECX2400 Homework 1

Zhiqi Fang

Student ID: 35954396

Question 1: Productivity and Working from Home

(i)

WFH may be correlated with u_t if exogeneity fails, that is, if $\text{cov}(WFH, u_t) \neq 0$. A concrete mechanism is omitted variable bias: because WFH is not randomly assigned, workers with higher unobserved motivation or self-discipline are more likely to select into WFH. Since these unobserved traits also increase productivity, so they enter the error term and make WFH correlated with u_t . Therefore, the observed relationship does not identify the causal effect *ceteris paribus*, and β_1 is upward biased.

(ii)

Since past productivity is observed before WFH is chosen, it is a valid pre-treatment control and can be included in a multiple regression. Controlling for past productivity helps reduce omitted variable bias when WFH is not randomly assigned. It captures prior differences across workers that would otherwise remain in the error term and be correlated with WFH. Therefore, the estimate of β_1 better captures the *ceteris paribus* effect of WFH.

(iii)

One confounder is time management ability, defined as a worker's pre- $t = 1$ capacity to organise tasks, meet deadlines, and work independently. Workers with stronger time management ability are more likely to be selected into WFH and are also more productive and they are also more productive, so if omitted it enters u_t and implies $\text{cov}(WFH, u_t) > 0$, leading to upward bias.

A second confounder is home internet speed, defined as the worker's pre-treatment internet bandwidth at home before $t = 1$. Faster home internet increases the feasibility of WFH and can also raise productivity by reducing communication delays and interruptions, so if omitted it enters u_t and implies $\text{cov}(WFH, u_t) > 0$, again leading to upward bias.

(iv)

We implement a two-sided t-tests on the coefficients of the two variables from part (iii) in the multiple regression. For each variable, the null hypothesis is $H_0 : \beta = 0$ and the alternative hypothesis is $H_1 : \beta \neq 0$, meaning the variable does not affect versus does affect productivity. The reasoning is that the standard error of each regression coefficient is used to calculate a t-statistic and test whether a particular explanatory variable is significantly related to the dependent variable, holding other factors constant. At $\alpha = 0.05$, reject H_0 if $|t| > 1.96$ or equivalently if the p-value is below 0.05; otherwise, fail to reject H_0 .

Question 2: Productivity and Flexible Hours

A

FlexHours may be correlated with the error term because Experience is omitted from the model, so it becomes part of v . Experience affects productivity and is also related to flexible hours, which creates omitted variable bias and violates $\text{cov}(FlexHours, v) = 0$. Therefore, α_1 reflects both the effect of FlexHours and part of the effect of Experience. Hence, α_1 is biased and cannot be interpreted as the *ceteris paribus* causal effect of flexible hours on productivity.

B

The omitted variable bias is positive.

Since experience raises productivity ($\gamma_1 > 0$) and is positively correlated with flexible hours ($\text{cov}(Experience, FlexHours) > 0$), and $\text{var}(FlexHours) > 0$.

Therefore, from $OV B = \gamma_1 \frac{\text{cov}(Experience, FlexHours)}{\text{var}(FlexHours)}$, it follows that $OV B > 0$.

C

The estimated α_1 is overestimated. Because $OV B > 0$, we have $\alpha_1 = \beta_1 + OV B > \beta_1$, so the estimated effect of flexible hours is upward biased and overstates the true causal effect. Intuitively, experience is positively correlated with both productivity and flexible hours, so flexible hours will capture part of the productivity effect that truly belongs to experience. Therefore, the coefficient on flexible hours is too large relative to the true *ceteris paribus* effect.

D

(i) Under random assignment, FlexHours is independent of both observed and unobserved factors affecting productivity, so the exogeneity condition $\text{cov}(FlexHours, u) = 0$ holds.

This means workers assigned to flexible and fixed hours are comparable on average before treatment, so differences in productivity can be attributed to the treatment rather than to pre-existing differences. Therefore, $\alpha_1 = \beta_1$, and the estimated coefficient identifies the true *ceteris paribus* causal effect of flexible hours on productivity.

(ii) Random assignment implies $\text{cov}(\text{Experience}, \text{FlexHours}) = 0$, because experience is a pre-treatment characteristic and is no longer related to treatment status. Hence, in the omitted variable bias formula, the covariance term becomes zero, so $OVB = 0$. As a result, Experience no longer biases the estimated effect of FlexHours, and omitted variable bias is eliminated.

E

No, under random assignment, the OLS estimator is already unbiased because FlexHours is independent of both observed and unobserved factors affecting productivity. Including additional controls such as education does not change this bias, since the exogeneity condition already holds. However, extra controls can shrink the variance of the error term and reduce standard errors, so they may improve the precision of the estimates. Therefore, additional controls affect precision, not bias.

F

(i) Self-selection into flexible working hours can bias the estimated α_1 because workers who choose flexible hours may already differ from other workers before treatment. For example, more motivated, more experienced, or better organised workers may be more likely to select into flexible hours, and these same pre-treatment characteristics also affect productivity. Therefore, the estimated effect of FlexHours mixes the treatment effect with pre-existing differences between workers, biasing α_1 .

(ii) A concrete labour-market example is that high-performing senior employees may be more likely to negotiate or be granted flexible hours than junior workers. In that case, workers with flexible hours may already have higher average productivity even before treatment. So part of the observed productivity difference reflects selection rather than the effect of flexible hours itself.

(iii) Comparing average productivity between workers with and without flexible hours may lead to incorrect conclusions about causality because a simple difference between two groups does not by itself identify a causal effect. The two groups may differ in pre-treatment characteristics, so the observed difference is not purely due to flexible hours. Therefore, higher average productivity among workers with flexible hours could reflect who selects into flexible hours rather than the true effect of flexible hours.

G

(i) Productivity could determine flexible hours when firms preferentially offer flexibility to high performers, or when productive workers self-select into flexible arrangements because they are trusted to work autonomously. Thus, productivity helps determine whether a worker receives or selects into flexible hours.

(ii) This creates a problem because the explanatory variable is no longer exogenous: flexible hours are partly determined by productivity, so $\text{cov}(FlexHours, u) \neq 0$. Consequently, the estimated coefficient on *FlexHours* does not isolate the causal effect of flexible hours on productivity. Instead, the regression mixes the effect of flexible hours with the effect of productivity feeding back into treatment assignment.

(iii) Reverse causality means the dependent variable affects the independent variable, so here productivity causes FlexHours. Omitted variable bias is distinct: it comes from an unobserved third variable (like experience or motivation) that influences both FlexHours and productivity. In both cases, the regressor is correlated with the error term, but the reason for the correlation is not the same.